



## The Keep-Alive Tap: An Interview Study of Locker Retention Under an Automated Inactivity-Reclamation Sweep

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**Abstract.** The University's Locker Retention Sweep reclaims any assigned storage locker unaccompanied by a logged tap of its entry sensor within a rolling 21-day window, reassigning the space to the next name on the waiting list. We report a qualitative study of locker holders the Sweep's own record credits as engaged despite negligible use recorded by a second, research-access-only door-open sensor the Sweep's holder-facing dashboard never surfaces. Semi-structured interviews with 46 flagged holders across the Recreation Centre and Library locker banks, thematically coded to saturation, sort what a tap without an opening is for into four types: a contingency reserve against imagined future need, a tap delegated to someone passing on the holder's behalf, a tap performed in distrust of the Sweep's own count, and a tap folded unnoticed into an unrelated daily route. The typology's distribution differs by bank, with contingency reserve concentrated at the Recreation Centre and distrust and folded-in routine concentrated at the Library. A term-long pilot doubling the Library's qualifying-tap threshold left the hidden sensor's genuine-open rate statistically unchanged. We read the tap less as evasion of the Sweep than as the Sweep's own proxy standing in, adequately by its own account, for a use it was never built to verify directly.

### Introduction

Any allocation rule that reclaims an underused resource needs a legible signal of use it can act on faster than the underlying use itself changes. The signal is rarely the thing itself: it is a proxy chosen for how cheaply it can be logged, audited, and acted on at scale. Airport slot allocation supplies a structural precedent at a much larger scale: mechanism-design work aims for egalitarian, congestion-aware, truthful slot allocation [1], one instance of a wider problem — allocating a scarce, time-sliced airport resource among competing claims — that recurs across other airport systems, surveyed for check-in counters [2]; whichever proxy a use-it-or-lose-it rule settles on, the rule and the proxy become, functionally, the same policy. Once a proxy is legible and the use behind it is not, the gap between the two is available to whoever finds it — not as fraud against a rule a holder intends to break, but as ordinary maintenance of a claim to

a resource they are reluctant to give up [3], [4]. The general form of the problem — a measure adopted as a target ceasing to describe the thing it was chosen to track — is well studied under Goodhart's law and its variants [5], [6], [7], [8], almost always in settings with an optimising agent gaming a formal score. A storage locker offers a smaller, more legible instance: a single tap, indistinguishable by the system from a genuine visit, resets the 21-day clock the University's Locker Retention Sweep otherwise runs against every assigned locker at the Recreation Centre and the Library.

We obtained research access to a second signal the Sweep's own dashboard never surfaces: a door-open sensor installed for facilities analytics and logged independently of the tap the Sweep's reclamation logic actually acts on. Comparing the two signals identifies locker holders the Sweep credits as engaged despite a door that, by the second sensor's account, rarely

opens — and interviews with 46 of them ask what the tap, absent the opening, is actually for.

Our contributions:

- we quantify, from a sensor the Sweep’s own policy does not consult, how far tap-based engagement over-credits use at both locker banks;
- we construct a four-part typology, from interviews coded to saturation, of what a keep-alive tap is for, and show its distribution differs by bank;
- our results from a threshold-doubling pilot suggest raising the qualifying-tap count changes little about how much a locker is actually opened.

## Related work

Territoriality research treats a claimed space as something people defend and retain independent of whether they still need it functionally, with psychological ownership persisting well past the point of instrumental use [3], [4]; hot-desking and non-territorial working policies carry their own measured costs for the employees who lose a fixed desk, from reduced self-assessed productivity to friction over shared amenities [9]. Use-it-or-lose-it allocation rules elsewhere supply a structural precedent: airport slot allocation has motivated mechanism-design work toward fairer, congestion-aware allocation [1], and the same problem — gating a scarce, time-sliced resource on a signal the system can act on at scale — recurs across other airport systems, surveyed for check-in counters [2]. Goodhart’s-law scholarship generalises the mechanism: once a measure is adopted as the criterion for an action, its relationship to the thing it was chosen to track degrades, a result formalised across several independent framings [5], [6], [7], [8]. Audit-culture research supplies the institutional register this paper works in: a measurement regime, once adopted, reorganises the behaviour of whoever it measures around satisfying the regime rather than the activity it was meant to track [10]. Building-occupancy research shows how hard genuine use is to observe directly at scale even with dedicated instrumentation — audio sensing [11], smart-meter inference [12], and calibrated simulation against Wi-Fi and badge data [13] all substitute an inferred signal for an occupancy nobody watches directly, the same substitution

a tap makes for a locker. Recent work on hoarding without exclusionary intent describes retention that answers imagined future need rather than any wish to keep a resource from someone else [14], a description this paper’s contingency-reserve category fits closely.

## Method

**Setting.** The University’s Locker Retention Sweep manages assigned lockers at two sites, the Recreation Centre (612 lockers) and the Library day-locker wall (284 lockers). Each locker’s entry reader logs a tap whenever a holder’s card is presented, regardless of whether the door is subsequently opened; the Sweep reclaims any locker without a logged tap in a rolling 21-day window, reassigning it in waiting-list order. A separate door-contact sensor, installed for facilities analytics and not consulted by the Sweep’s reclamation logic, independently logs genuine opens; its existence is not disclosed on the Sweep’s holder-facing dashboard.

**Sample.** Facilities provided, under a standard research data-sharing arrangement, a de-identified list of holders whose tap record kept them above the 21-day threshold in each of the past two terms while their door-open count over the same window fell in the bottom decile for their site. We invited a stratified random sample of 70 flagged holders, proportional to each site’s flagged population, and interviewed the 46 who consented (27 Recreation Centre, 19 Library; 66% response).

**Interviews.** Each interview ran 20–30 minutes, opening from a fixed prompt — “walk me through the last time you tapped in without opening the locker” — and audio-recorded with consent. Two researchers independently coded transcripts using thematic analysis [15], open-coding the first third of the sample before agreeing a shared codebook and applying it to the remainder; recruitment and coding continued until three consecutive interviews introduced no code not already in the book.

**Triangulation.** For each interviewee, we compared the coded account against that holder’s own tap-and-open log over the preceding term, checking only whether the pattern the log showed was compatible with the account given, not whether the account could be independently confirmed in full.

**Pilot ablation.** Partway through data collection, Facilities piloted a doubled qualifying-tap requirement at the Library site only (two taps per fortnight rather than one), leaving the Recreation Centre’s threshold unchanged as a concurrent control. We treat this as a natural ablation on the Sweep’s own policy variable and compare each site’s door-sensor-logged open rate for the term before and the term during the pilot.

**Reliability.** Inter-rater agreement on the four-part typology was assessed with Cohen’s kappa [16],

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where  $p_o$  is the observed proportion of coded segments on which both coders agreed and  $p_e$  is the proportion expected by chance given each coder’s marginal category rates. The typology reached kappa = 0.81 across the full sample.

## Results

**Tap-based engagement over-credited use at both sites.** Among the 46 interviewed holders, the door-open sensor recorded a median 0.3 genuine opens per fortnight against the Sweep’s tap record of full engagement; 39 of 46 (85%) opened their locker on fewer than two occasions across the entire term preceding interview despite never once falling foul of the 21-day tap window. Table 1 summarises recruitment and response by site.

| Site              | Invited   | Interviewed | Response   |
|-------------------|-----------|-------------|------------|
| Recreation Centre | 46        | 27          | 59%        |
| Library           | 24        | 19          | 79%        |
| <b>Total</b>      | <b>70</b> | <b>46</b>   | <b>66%</b> |

Table 1: Recruitment and response by site, drawn from the Sweep’s own flagged-holder list.

**Coding sorted every account into one of four types.** Coding to saturation across the full sample assigned every keep-alive-tap account to one of four types: contingency reserve, in which a holder retains the locker against an imagined future need with no current use to speak of; delegated tapping, in which someone else — a training partner, a housemate — taps the reader on the holder’s behalf while passing; distrust of the sweep, in which a holder taps more often than the stated rule requires because they do not trust the count to register correctly; and

folded-in routine, in which the tap survives as an unconscious step in an unrelated daily path, with no locker use attached to it at all. No fifth type emerged after the thirty-first interview, and none of the last twelve interviews introduced an account the codebook could not already accommodate.

**The typology’s distribution differs sharply by site.** Figure 1 reports the share of coded segments assigned to each type, split by locker bank. The Recreation Centre concentrates in contingency reserve (41%) and delegated tapping (22%); the Library concentrates instead in distrust of the sweep (32%) and folded-in routine (31%), with contingency reserve falling to 26% and delegated tapping to 11%.

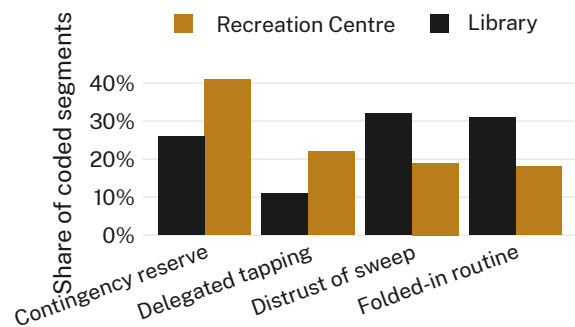


Figure 1: Share of coded interview segments by keep-alive-tap type and locker bank: the Recreation Centre clusters toward contingency reserve and delegated tapping, the Library toward distrust of the sweep and folded-in routine.

**Doubling the qualifying-tap threshold left genuine use unchanged.** Figure 2 compares the door-sensor-logged open rate at each site for the term before and the term during the Library’s doubled-threshold pilot. The Recreation Centre’s control distribution is effectively unmoved (median 0.31 before, 0.30 during); the Library’s pilot distribution shifts marginally in the opposite direction from what a working deterrent would predict (median 0.34 before, 0.37 during;  $p = 0.61$ , n.s.).

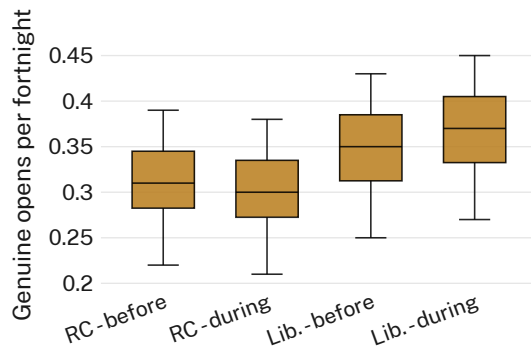


Figure 2: Genuine opens per fortnight, by site and pilot period: the Library’s doubled qualifying-tap threshold left the open rate statistically indistinguishable from its own pre-pilot term and from the Recreation Centre’s unchanged control.

## Discussion

The Sweep’s tap, cheap to log and immediate to act on, stands in adequately for use only when a holder is indifferent to being reclaimed; every account we coded describes a holder who is not. Framed through Goodhart’s law, the tap was never the thing the Sweep wanted to track, but once holding a locker turned on the tap alone, the tap displaced the opening as the object worth managing [5], [7]. The doubled-threshold pilot’s null result at the Library suggests the substitution is close to complete: raising the cost of the proxy by exactly the amount the policy varies does not restore any relationship between the proxy and the door it was meant to stand in for. The contingency-reserve category, the largest single type at the Recreation Centre, reads less as an attempt to deny the locker to anyone on the waiting list than as retention against an imagined future need with no exclusionary intent behind it, consistent with recent hoarding research that separates the two [14]. Territoriality accounts of a claimed space outlasting its functional need [3], [4] describe the same holder from the ownership side of the behaviour this paper describes from the measurement side.

## Limitations

Our sample is drawn entirely from holders the Sweep itself flagged as engaged; we cannot characterise how holders whose taps and opens already agree would describe their own behaviour, though the four-part typology’s stability across two sites under a shared policy but different physical layouts is some evidence the

categories are not an artefact of a single site’s culture. The pilot ablation ran for a single term at one site rather than a randomised multi-site trial, though the concurrent, untouched control at the Recreation Centre rules out a University-wide seasonal explanation for the null result. Motive is self-reported throughout; we treat consistency with each holder’s own tap-and-open log as evidence of compatibility, not as independent confirmation of what a holder actually intended.

## Conclusion

Interviews with 46 locker holders the University’s Locker Retention Sweep credits as engaged, despite a door-open sensor its own policy never consults, sort the keep-alive tap into four types — contingency reserve, delegated tapping, distrust of the count, and folded-in routine — distributed differently at the Recreation Centre and the Library. A term-long pilot doubling the Library’s qualifying-tap threshold left genuine opens statistically unchanged, suggesting the tap has substituted for the opening more completely than a heavier version of the same proxy can undo. We take no position on whether the Sweep’s threshold should change; that is a question for whoever administers the waiting list, not for a study built to describe what the tap has come to mean. Future work might follow a flagged holder across several reclamation cycles to see how long a kept-alive locker stays kept alive before someone lets it go.

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